LEX PROGRAM TO ADD LINE NUMBERS TO A FILE

1. AIM

To write and execute a Lex program that reads an input file and displays its contents on standard output with a line number added to every line.

2. PSEUDOCODE

1. Start the program and initialize the line number to 1.
2. Open the input file in read mode.
3. If the file cannot be opened, display an error and stop.
4. Set the Lex input stream to the opened file.
5. For each line of text, print the current line number followed by the line contents.
6. Increment the line number whenever a newline is encountered.
7. Close the file and stop.

3. LEX PROGRAM

```
%option noyywrap

%{
#include <stdio.h>

int line_no = 1;
%}

%%
\n      { line_no++; }
[^\n]+  { printf("%d: %s\n", line_no, yytext); }
%%

int main(void)
{
    FILE *fp = fopen("input.txt", "r");

    if (fp == NULL)
    {
        printf("Error: Cannot open input.txt\n");
        return 1;
    }

    yyin = fp;
    yylex();

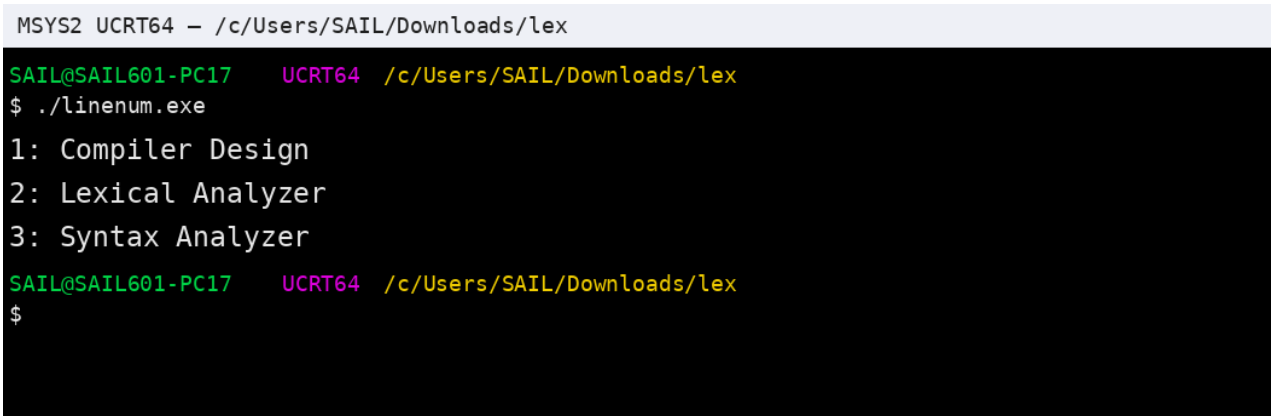
    fclose(fp);
    return 0;
}
```

4. SAMPLE INPUT FILE (input.txt)

```
Compiler Design
Lexical Analyzer
Syntax Analyzer
```

5. OUTPUT IMAGE

Illustrative expected output for the sample input. Replace this illustration with a screenshot of your actual terminal output if your lab submission requires evidence of execution.



6. RESULT

Thus, the Lex program to read an input file and display its contents with line numbers was prepared. For the sample input, the expected output numbers the three lines as 1, 2, and 3.